

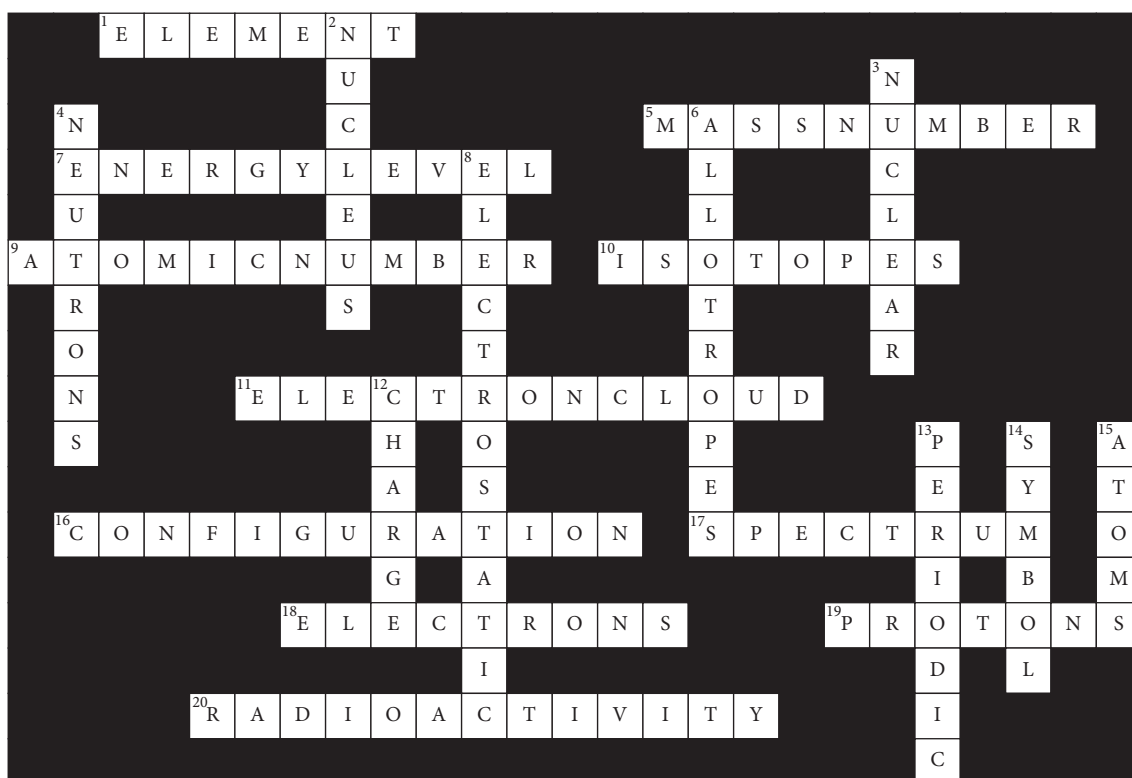
ACTIVITY SHEET ANSWERS

Chapter 1 Suggested answers

Prior knowledge

- | | | |
|--|---------------------------------|-----------------------|
| 1 atoms | 7 speed | 14 radioactivity |
| 2 nucleus | 8 positive; negative; uncharged | 15 shells; least |
| 3 protons, neutrons,
electrons (in any order) | 9 protons | 16 fixed; two; eight |
| 4 heavier | 10 equal; six | 17 full; share |
| 5 lighter | 11 element; protons | 18 periodic |
| 6 mass | 12 symbol | 19 periods; groups |
| | 13 neutrons; isotopes | 20 chemical; occupied |

Activity sheet 1.1



Activity sheet 1.3

- 1 Americium-241 spontaneously decays into neptunium-237 and an alpha particle.
- 2 That the chamber contains a source of radioactivity
- 3 $^{241}_{95}\text{Am}$ has 95 protons and 146 neutrons; $^{237}_{93}\text{Np}$ has 93 protons and 144 neutrons
- 4 2 neutrons and 2 protons; because the neptunium isotope has two less protons and two less neutrons than the americium isotope
- 5 Turns them into ions by removing electrons from them
- 6 To create an electrical field/an electrical potential difference
- 7 The ionised air or smoke molecules are charged and will move in response to the electric field, thus creating an electrical current.
- 8 As they are heavier molecules, they do not accelerate as quickly as the ionised air molecules, which reduces the strength of the current.
- 9 That the battery is getting flat and needs replacing
- 10 Because they will only travel for a few centimetres before they are neutralised by air molecules (They become inert helium atoms, ^4_2He .)

Activity sheet 1.4

Suggested colour coding:

GROUP	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
PERIOD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1,00794 H HYDROGEN																	4,002602 He HELIUM
2	6,941 Li LITHIUM	9,01218 Be BERYLLIUM																20,1797 Ne NEON
3	22,989769 Na SODIUM	24,305 Mg MAGNESIUM											26,981538 Al ALUMINUM	28,0855 Si SILICON	30,973761 P PHOSPHORUS	32,065 S SULFUR	35,4527 Cl CHLORINE	39,948 Ar ARGON
4	39,0983 K POTASSIUM	40,078 Ca CALCIUM	44,955911 Sc SCANDIUM	47,867 Ti TITANIUM	50,9415 V VANADIUM	51,9961 Cr CHROMIUM	54,938044 Mn MANGANESE	55,845 Fe FERRUM	58,9332 Co COBALT	58,9332 Ni NICKEL	63,546 Cu COPPER	65,409 Zn ZINC	69,723 Ga GALLIUM	72,64 Ge GERMANIUM	74,92160 As ARSENIC	78,96 Se SELENIUM	79,904 Br BROMINE	83,798 Kr KRYPTON
5	85,4678 Rb RUBIDIUM	87,62 Sr STRONTIUM	88,90584 Y YTRIUM	91,224 Zr ZIRCONIUM	92,90638 Nb NIOBIUM	95,94 Mo MOLYBDENUM	97,905 Tc TECHNETIUM	101,07 Ru RUTHENIUM	102,9055 Rh RHODIUM	106,42 Pd PALLADIUM	107,8682 Ag SILVER	112,411 Cd CADMIUM	114,818 In INDIUM	118,710 Sn TIN	121,760 Sb ANTIMONY	127,60 Te TELLURIUM	126,90447 I IODINE	131,293 Xe XENON
6	132,90545 Cs CESIUM	137,327 Ba BARIUM	138,90547 La - Lu LANTHANIDE	174,967 Hf HAFNIUM	180,94788 Ta TANTALUM	183,84 W TUNGSTEN	186,207 Re RHENIUM	186,207 Os OSMIUM	190,23 Ir IRIDIUM	195,078 Pt PLATINUM	196,96655 Au GOLD	200,59 Hg MERCURY	204,3833 Tl THALLIUM	207,2 Pb LEAD	208,98038 Bi BISMUTH	209 Po POLONIUM	210 At ASTATINE	222 Rn RADON
7	223,0197307 Fr FRANCIUM	226 Ra RADIUM	227 Ac - Lr ACTINIDE	261 Rf RUTHERFORDIUM	268 Db DUBNIUM	271 Sg SEABORGIUM	272 Bh BOHRILIUM	277 Hs HASSIUM	277 Mt MEITNERIUM	281 Ds DARMSTADTIUM	281 Rg ROENTGENIUM	285 Cn COFERNITIUM	289 Fl FLEROVIUM	289 Lv LIVERMORIUM				

Aaa - Gas

Aaa - Solid

Aaa - Liquid

Aaa - Synthetic

LANTHANIDE

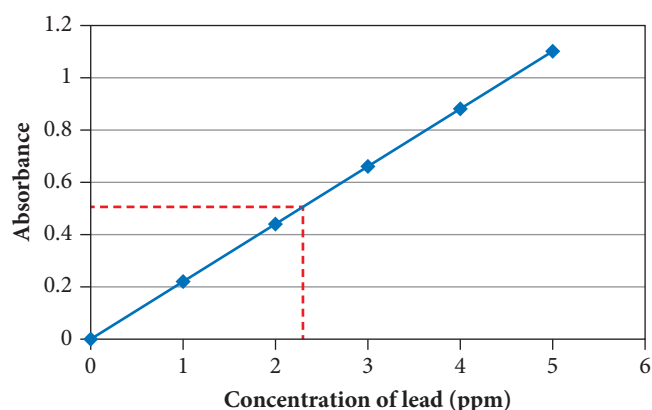
57 138,905 La LANTHANUM	58 140,116 Ce CERIUM	59 140,90765 Pr PRASEODYMIUM	60 144,242 Nd NEODYMIUM	61 145 Pm PROMETHIUM	62 150,36 Sm SAMARIUM	63 151,964 Eu EUROPIUM	64 157,25 Gd GADOLINIUM	65 158,92534 Tb TERBIUM	66 162,500 Dy DYSPROSIUM	67 164,9303 Ho HOLMIUM	68 167,259 Er ERBIUM	69 168,934 Tm THULIUM	70 173,04 Yb YTTERIUM	71 174,967 Lu LUTETIUM
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ACTINIDE

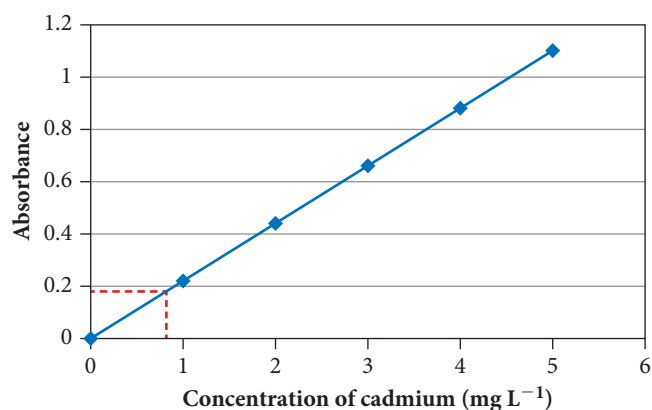
89 227 Ac ACTINIUM	90 232,0381 Th THORIUM	91 231,03688 Pa PROTACTINIUM	92 238,0289 U URANIUM	93 237 Np NEPTUNIUM	94 244,06 Pu PLUTONIUM	95 243 Am AMERICIUM	96 247 Cm CURIUM	97 247 Bk BERKELIUM	98 251 Cf CALIFORNIUM	99 252 Es EINSTEINIUM	100 257 Fm FERMIUM	101 258 Md MENDELEVIUM	102 259 No NOBELIUM	103 262 Lr LAWRENCIUM
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Activity sheet 1.5

1 Calibration curve for determining the concentration of lead



2 Calibration curve for determining the concentration of cadmium



Chapter 1 Revision

1 a → G, b → X, c → C, d → Q, e → U, f → V, g → F, h → J, i → P, j → A, k → K, l → M, m → E, n → N, o → H, p → T, q → D, r → O, s → B, t → R, u → L, v → S

- 2 a Atomic mass is the number of protons in the nucleus. Mass number is the number of protons and neutrons in the nucleus.
- b Isotopes are atoms of an element with different numbers of neutrons. Allotropes are different physical forms of an element.
- c A period is a row of the periodic table. A group is a column of the periodic table.
- d Atomic absorption spectroscopy is used to determine the concentration of an element. Mass spectrometry is used to determine the relative abundance of different isotopes.

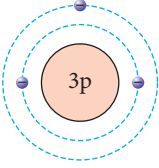
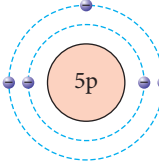
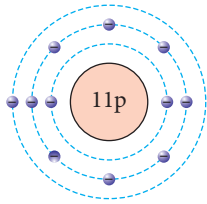
3	Particle	Symbol	Charge	Location	Relative Mass
	proton	p	+1	nucleus	1
	electron	e	-1	electron cloud	1/1800
	neutron	n	0	nucleus	1

	Atom A	Atom B	Atom C
number of protons	5	9	18
number of electrons	5	9	18
number of neutrons (N)	6	10	22
atomic number (A)	5	9	18
mass number (Z)	11	19	40
element symbol	B	F	Ar
atomic representation (A_ZX)	${}^{11}_5\text{B}$	${}^{19}_9\text{F}$	${}^{40}_{18}\text{Ar}$

- 5 Electrons are held in the electron cloud around the nucleus by electrostatic attraction between the electrons and the protons in the nucleus. Particles in the nucleus are held together by the strong nuclear force.
- 6 The atoms of non-naturally occurring elements tend to break down rapidly because the forces inside their nuclei are not balanced, which makes the atoms unstable.

- 7 a 17 protons
- b 20 neutrons in the left atom and 18 neutrons in the right atom
- c Both have 17 protons, making them Cl atoms, but the differing numbers of neutrons makes them isotopes
- d $(76 \times 35) + (24 \times 37)/100 = 35.4$

- 8 a 2 in first shell, 8 in the second, and 8 or 18 in the third shell
- b Electrons fill shells from the innermost shell outwards.
- c Electrons in the innermost shells have the least amounts of energy; those on the outermost shells have the most amounts of energy.

Lithium	Boron	Sodium
		
2, 1 (or $1s^2, 2s^1$)	2, 2, 1 (or $1s^2, 2s^2, 2p^1$)	2, 8, 1 (or $1s^2, 2s^2, 2p^6, 3s^1$)

- e Valence electrons are electrons in the outermost occupied shell of an atom.
- 9 a Atomic radii decrease across periods (from left to right) but increase down groups.
- b Electronegativities increase across periods (from left to right) but decrease down groups.
- c Ionisation energies increase across periods (from left to right) but decrease down groups.
- d Electron affinities increase across periods (from left to right) but decrease down groups.
- 10 a An atom is in its ground state when all of its electrons are in the lowest possible energy levels. An atom is in an excited state when some of its electrons are not in the lowest possible energy levels.
- b If an electron absorbs a suitable amount of energy it will move into a higher energy level for a brief period of time before releasing that energy as light of a particular frequency as it falls back to its original lower energy level.

- c Each brightly coloured line represents the energy released as a particular electron 'falls' from a higher energy level to a lower energy level.
- d The shells of the two elements have differing energy levels, so different wavelengths of light are emitted as excited electrons return to their ground states.